



Medication Management Procedure

Clozapine

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	All areas that deal with inpatients.	Medical, Nursing and Pharmacy

Introduction

This procedure serves to compliment the Department of Health document “Guidelines for the Safe and Quality Use of Clozapine Therapy in the WA health system” and assist in the management of clozapine in a non-mental health setting such as SCGOPHCG.

Clozapine is a high risk medication due to the risk of potentially fatal side effects. Mismanagement of clozapine, especially during initiation or after more than 48 hours of missed doses, increases the risk of these side effects occurring and has resulted in increased lengths of stay and in some cases death. Vigilant monitoring is required for patients being treated with clozapine, with the Therapeutic Goods Administration implementing mandatory haematological monitoring standards in Australia to minimise the risk of clozapine side effects.

Risk Statement

Non-compliance with this procedure will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

Definitions

ClopineCentral™	The TGA-mandated monitoring system administered by Pfizer Pty Ltd. It requires all patients, prescribing doctors, dispensing pharmacists, centre co-ordinators and registered centres using clozapine (Clopine®) to be registered with the database. At the time of procedure development, Pfizer Pty Ltd holds the public hospital tender for clozapine in Western Australia.
ClopineCentral™ blood test	White Cell and Neutrophil counts that must be performed at regular intervals for patients taking clozapine, as per the ClopineCentral™ protocol.



ClopineCentral™ Protocol	A set of mandated requirements that must be met for the prescribing and dispensing of clozapine at a registered centre. The ClopineCentral™ Protocol is the overarching document maintained by Pfizer Pty Ltd that details all procedures, forms and contact information relating to the management of the Clopine® brand of clozapine. See Appendix 2 for details on how to access the protocol.
Registered Centre	A hospital, clinic or other facility that is involved with the use of clozapine, in conjunction with ClopineCentral™. Patients are listed as 'belonging' to a specific centre through their registration at the centre.
Centre Co-ordinator	Nominated person who is responsible for ensuring the registered centre operates in accordance with the ClopineCentral™ protocol. This person for SCGH is Inpatient Pharmacy Pharmacist in Charge. This person for OPH is the Dispensary Manager.
CLP	Clinical Liaison Psychiatrist
POC	Psychiatrist on call

Procedure

1.0 Continuation of Clozapine Therapy for an Inpatient

DEATHS HAVE OCCURRED AS A RESULT OF THE MISMANAGEMENT OF CLOZAPINE.

BEFORE PRESCRIBING/ADMINISTERING/WITHHOLDING/MODIFYING CLOZAPINE:

- OBTAIN COMPULSORY INFORMATION IN TABLE 1
- CONTACT:
CLP/POC OR EMERGENCY DEPARTMENT PSYCHIATRY TEAM (depending on scenario)
AND
YOUR CLINICAL PHARMACIST AT THE NEAREST OPPORTUNITY (IF APPLICABLE)
- Clozapine therapy should not be modified or ceased before contacting the CLP or POC.

SEE APPENDIX 1 FOR A FLOWCHART OF THE CLOZAPINE ADMISSION PROCESS



1.1 Obtaining Compulsory Information

- The compulsory information in Table 1 must be obtained **before** any decisions regarding clozapine are made including administration and withholding.
 - Compulsory information must be obtained and documented appropriately in medical record for all admitted patients taking clozapine regardless of circumstance.
 - This must be done as soon as practical and must be treated as a priority.
 - Clinical judgement must be used when using historical documentation to obtain the compulsory information as it may be outdated.
 - If the compulsory information cannot be obtained, see Section 1.2.
- ClopineCentral should be contacted to collect compulsory information as it is a central repository of information maintained for all patients taking clozapine. This information is available over the phone, patients do not need to be transferred to the SCGH Registered Centre.
 - During AEST business hours call 1800 656 403.
 - The ClopineCentral call centre is not available out of AEST business hours or during New South Wales public holidays. In this case, call the next business day and use local sources of information in the interim.
 - If deemed to be urgent, an after-hours agent is available for emergency cases. The relevant contact number is available at the end of the ClopineCentral after-hours answering machine message.

	INFORMATION	POTENTIAL SOURCES	RATIONALE
COMPULSORY INFORMATION	Current dose	• Patient/relative/carer • Patient's own medicine • Pharmacy/clinic/hospital, • Dose administration aid (Webster Pak®/pre-packed sachet), • Previous admission documents (use clinical judgement) • ClopineCentral™	• Ascertain correct dose to administer, if applicable
	Date and time of last dose including compliance history	• Patient/relative/carer • Dose administration aid (Webster Pak®/pre-packed sachet; use clinical judgement) • Hospital transfer documentation (use clinical judgement)	• To ascertain dose history and ensure therapy is not restarted suddenly if it has been more than 48 hours between doses.
	Results of last ClopineCentral™ White Cell and Neutrophil Count (if unavailable, use inpatient blood tests that are less than 48 hours old)	• ClopineCentral™ • Pharmacy/clinic/hospital • Inpatient test results (<48 hours old)	• To screen for potential clozapine side effects (neutropenia)
	Smoking status: • Cigarettes/day • Smoking history	• Patient/relative/carer • Previous admission documents (use clinical judgement)	• Smoking induces the metabolism of clozapine. • Ceasing smoking during admission causes a spike in clozapine serum concentration, increasing risk of life-threatening side effects
	Concurrent medications (particularly CYP1A2/CYP2D6 inhibitors/inducers; see Appendix 3 for examples)	• Patient/relative/carer • Dose administration aid (Webster Pak®/pre-packed sachet; use clinical judgement) • Hospital transfer documentation (use clinical judgement)	• If medications are to be ceased and they interact with clozapine, the CLP must be contacted in order to give appropriate advice regarding monitoring and dosing etc.
DESIRABLE INFORMATION	ClopineCentral™ Blood test interval	• Patient/relative/carer • ClopineCentral™ • Pharmacy/clinic/hospital	• Helps to establish if the patient is in titration or maintenance phase.
	Due date of next ClopineCentral™ blood test	• Patient/relative/carer • ClopineCentral™ • Pharmacy/clinic/hospital	• To ascertain if a blood test will need to be recorded during the patient's admission.
	ClopineCentral™ registration number	• ClopineCentral™ • Pharmacy/clinic/hospital	• Makes certain actions in ClopineCentral™ easier to complete.

Table 1. Mandatory and desirable information required before charting/administering/withholding clozapine.

1.2 When to Contact the CLP/POC

- If any of the following is present, **the CLP/POC must be contacted as a priority** for advice before any decision regarding clozapine is made by the treating medical team:
 - Any compulsory information as per Table 1 is unknown, unclear or in doubt.
 - Patient is a smoker.
 - Blood dyscrasias.
 - Severe constipation.
 - Signs and symptoms of clozapine toxicity.
 - Patient has missed ≥ 48 hours of doses, or is non-compliant, INCLUDING irregular dosing.
 - Risk of or current hepatic impairment.
- If any information is incomplete or in doubt, or the CLP/POC cannot be contacted, wait until suitable advice is obtained from the Psychiatry Team. Do not prescribe or administer clozapine in these circumstances until the CLP/POC has given advice.

1.3 Prescribing Clozapine and Ongoing Patient Monitoring Requirements

- If all compulsory information is available and none of the points in 1.2 are present:
 - Notify the Emergency Department Psychiatric Team of the patient to flag them within the Psychiatry Team.
 - If the ward/area has a clinical pharmacy service, the nursing or medical team must notify the relevant clinical pharmacist of every clozapine patient as a priority. This includes patients who are new admissions, hospital-hospital transfers, or internal transfers.
 - Treating team to prescribe clozapine on the WA Hospital Medication Chart (WA HMC).
 - Complete a formal referral to the CLP regardless of reason for admission.
- All patients receiving clozapine must have white blood cell and neutrophil counts performed in accordance with the ClopineCentral™ Protocol. Monitoring requirements should continue upon admission to SCGH.

1.4 When to Transfer a Patient to the SCGH Registered Centre

- Transferring a patient to the SCGH or OPH Registered Centre (“transferring”) is only performed for patients:
 - Who have had their clozapine therapy ceased.
 - Who may have their clozapine dose changed, including re-titration.
 - Where it is anticipated that a ClopineCentral™-mandated blood test will be due to be performed during their admission.
 - Who have evidence of clozapine toxicity.
 - If clozapine is dispensed for them on discharge.
- The above practice is preferred as patients do not have to be transferred to another Registered Centre upon discharge. This simplifies the associated administrative process, the patient’s transition to the community, and ensures continuity of therapy.
- The ward-based clinical pharmacist associated with the patient’s care is responsible for

transferring the patient. During weekends/public holidays, this can be done the next business day. The pharmacist transferring the patient must be registered with ClopineCentral.

- If the patient is in an area where there is no clinical pharmacy service, contact the Inpatient Pharmacy Pharmacist in Charge (SCGH) or the Dispensary Manager (OPH).
- To transfer a patient:
 - During AEST business hours: call ClopineCentral™ on 1800 656 403.
 - Outside of AEST business hours/New South Wales public holidays, wait until the next east coast business day. Patients can be transferred in retrospect. It is not vital that this process be completed immediately, however it should be done at the nearest available opportunity.

1.5 Supply of Clozapine

- Supply of clozapine will only occur once the steps above have been completed. Supply of clozapine from SCGH/OPH Pharmacy will be on an individual patient basis and will only occur after pharmacist consultation.
- Products available at SCGH and OPH are:
 - Tablets: 25mg, 50mg, 100mg and 200mg
 - Liquid: 50mg/mL
- INPATIENT SUPPLY – SCGH
 - Patient's own medicine should be used whenever possible.
 - If medication is to be supplied from SCGH Pharmacy, consult your ward-based clinical pharmacist for further information. If your area does not have a clinical pharmacy service, contact the Inpatient Pharmacy Pharmacist In-Charge on (08) 6457 1114.
 - All orders for clozapine must have WCC and neutrophil results annotated by the pharmacist. If there is no clinical pharmacy service, this information will be checked and annotated on the order by the checking pharmacist in Inpatient Pharmacy. Blood test results must be less than 48 hours old. Supply of clozapine will not occur in the absence of appropriate blood test results.
 - If supply of clozapine is required after hours, a staff member directly involved with the patient's care must contact the POC/Emergency Department Psychiatry team (whichever is applicable) **prior** to contacting the on-call pharmacist. Any callers who have not done this will be referred to do so by the on-call pharmacist before supply is authorised.
 -
- INPATIENT SUPPLY – OPH
 - Consult your ward-based clinical pharmacist for further information on obtaining supply from OPH pharmacy.
 - All orders for clozapine must have WCC and neutrophil results annotated by the pharmacist. If there is no clinical pharmacy service, this information will be checked and annotated on the order by the checking pharmacist in Inpatient Pharmacy. Blood test results must be less than 48 hours old. Supply of clozapine will not occur in the absence of appropriate blood test results.

- If supply is required after hours, the AHNM may be contacted to obtain supply after satisfying all compulsory information relating to the patient/dose (above).
- **DISCHARGE SUPPLIES**
 - If an admission was not associated with the patient's clozapine therapy and no changes were made to the clozapine dose, it must not be supplied by SCGH on discharge unless circumstances are extenuating.
 - If it is deemed that supply is required on discharge, the patient should be directed to obtain supply from the pharmacy that usually supplies them with clozapine in the community or a community pharmacy that can do so, e.g. if the patient lives in a rural area and is staying in the metropolitan area. A handover must be given to the relevant community pharmacy in this scenario.
 - If extenuating circumstances dictate that clozapine must be supplied by SCGH, this will only be done after other mechanisms of supply (patient's usual pharmacy, supply at home etc.) have been exhausted.
 - The CLP or Emergency Department Psychiatric Team (after hours only) must be contacted as soon as this is identified to avoid delays on discharge.
 - Prescriptions must only be written by a prescriber who is registered with ClopineCentral™ to guarantee the supply can be appropriately documented as per the ClopineCentral™ Protocol.
 - The Inpatient Pharmacy Pharmacist In-Charge who is registered with ClopineCentral™ must be satisfied that:
 - The prescription is safe and appropriate.
 - The prescription is signed by a medical officer who is a member of the CLP or Emergency Department Psychiatric Team and registered with ClopineCentral™.
 - The prescription is accompanied by the results of a white cell count and neutrophil count less than 48 hours old.
 - When dispensing the clozapine prescription, the Inpatient Pharmacy Pharmacist In-Charge **must**:
 - Transfer the patient to the SCGH Registered Centre. See Section 1.4.
 - Record the dispensing in ClopineCentral™. See ClopineCentral™ Protocol for further information.
 - The quantity supplied must not exceed sufficient quantity to last until the patient's next due blood test.
 - Note: If registration of prescriber and/or pharmacist is required, the documentation of the dispensing in ClopineCentral can be completed in retrospect. Registration of staff should not delay supply if the pharmacist is confident that it is appropriate and all required information is present.

2.0 Initiation of Clozapine Therapy for an Inpatient

- Clozapine must not be initiated without formal referral to the CLP.
- Clozapine can only be initiated under the direct supervision of a consultant psychiatrist.
- Initiation of clozapine must be prescribed on the WA Adult Clozapine Initiation and Titration Chart available from Inpatient Pharmacy.
- The monitoring requirements detailed on the front and inside of the chart must be adhered to by medical and nursing staff.
- The CLP must contact the Centre Co-ordinator to facilitate the registration of the patient.

3.0 Missed Doses and Re-Titration of an Inpatient Currently Treated With Clozapine



RECOMMENCING CLOZAPINE WITHOUT DOSE TITRATION
HAS BEEN ASSOCIATED WITH TOXICITY AND DEATH

**DO NOT ADMINISTER ANY DOSES UNTIL THE CLP/POC
HAS BEEN CONSULTED AND
ADVICE HAS BEEN RECEIVED.**

- If more than 48 hours has passed since the last dose was taken by the patient, dose titration may be required when re-commencing.

Refer to the CLP/POC immediately.

- Re-titration should only occur under direct supervision of a consultant psychiatrist.
- Re-titration of clozapine must be prescribed on the WA Adult Clozapine Initiation and Titration Chart.
- The monitoring requirements detailed on the front and inside of the chart must be adhered to by medical and nursing staff.
- Re-titration may occur on a general ward but must be under the direct supervision and guidance of a consultant psychiatrist.
- The patient must be transferred to the SCGH or OPH Registered Centre to document interruption to therapy and subsequent re-titration if applicable. See Section 1.4.
- ClopineCentral must be notified of an interruption to clozapine therapy. This can be done via:
 - A phone call to ClopineCentral during AEST business hours by the CLP. This can be done in retrospect if outside of AEST business hours.
 - A 'Therapy Event Form' submitted to ClopineCentral™ by the treating consultant psychiatrist, or designated person within the CLP.
- Haematological monitoring requirements will change depending on the patient's current monitoring frequency, as per the ClopineCentral™ Protocol.

4.0 Interruption of Clozapine Therapy

- The CLP/POC must be consulted if there is an interruption to clozapine therapy.
- ClopineCentral™ must be notified of any therapy interruption by phone.
- Haematological monitoring requirements will change depending on the patient's current monitoring frequency. Refer to the WA Adult Clozapine Initiation and Titration Chart.

5.0 Discontinuation of Clozapine Therapy

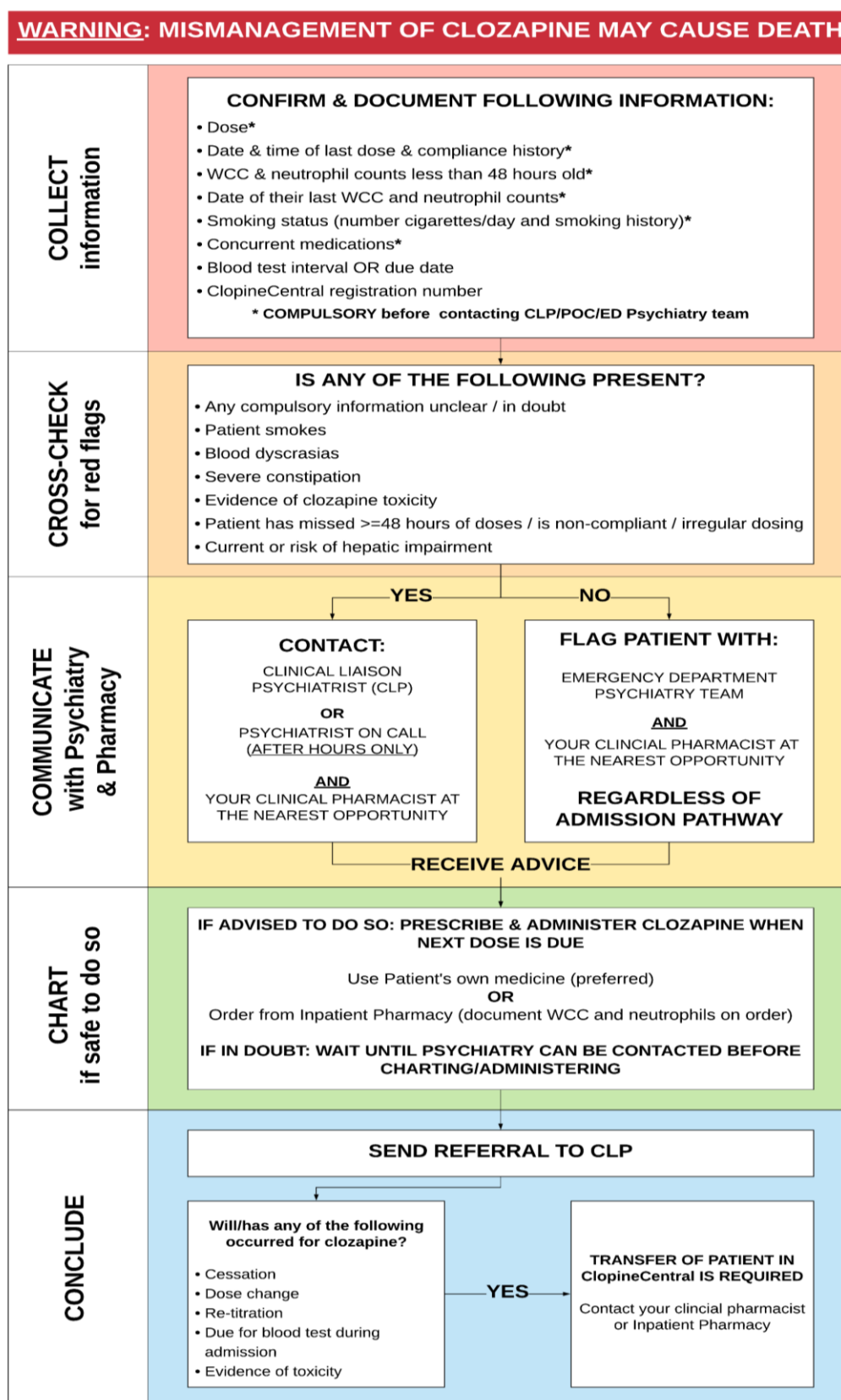
- The decision to cease therapy must always be made in consultation with the CLP/POC. Relapse following abrupt cessation of clozapine can be refractory to treatment. Side effects of clozapine are not necessarily a contraindication to continued treatment.
- Monitoring of haematological parameters is required after cessation of clozapine. Frequency and duration depend on monitoring requirements in place at the time of discontinuation.
- A blood test is required to assess full blood picture at cessation of therapy. See ClopineCentral™ Protocol for further information.
- In the event of cessation of clozapine therapy ClopineCentral™ must be notified. See ClopineCentral™ Protocol for further information.

6.0 Transfer of a Patient to Another Registered Centre

- When a patient undergoing clozapine therapy is discharged or transferred to another facility, the patient must be transferred to the Registered Centre that will be taking over the patient's care. This only occurs if the patient was initially transferred to the SCGH/OPH Registered Centre.
- The treating medical team must ensure that they clearly document the details of the ClopineCentral™ Centre who will be taking over the patients care in the inpatient notes.
- The Centre Co-ordinator must be informed of the discharge or transfer to facilitate transfer in ClopineCentral™ by the treating medical team or clinical pharmacist caring for the patient.

Related Documents

Appendix 1. Clozapine Admission Flowchart



Appendix 2. Accessing the ClopineCentral™ Protocol

- Visit <https://www.pfizermedicalinformation.com.au/en-au/clopine/>
- Register your personal details to obtain a username and password. **NB:** There may be a delay in the activation of your account as healthcare practitioner details must be verified with AHPRA.
- Log in using your credentials.
- Follow the links to access information such as:
 - The ClopineCentral™ protocol.
 - ClopineCentral™ registration forms.
 - Patient-related forms.
 - Product information.

Appendix 3. CYP1A2 and CYP2D6 Inducers/Inhibitors

	CYP1A2	CYP2D6
Inducers (reduces clozapine levels)	Omeprazole Phenobarbital Phenytoin Rifampicin Ritonavir Tobacco smoking	N/A
Inhibitors (increases clozapine levels)	Cannabidiol Ciprofloxacin Combined oral contraceptives Fluvoxamine Peginterferon x-2a Vemurafenib Verapamil	Abiraterone Amiodarone Bupropion Celecoxib Cinacalcet Cobicistat Duloxetine Fluoxetine Methadone Mirabegron Paroxetine Terbinafine
NB: THIS LIST IS NOT EXHAUSTIVE – consult external references		

Source: Australian Medicines Handbook (online), accessed 27 July 2023.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [WA Adult Clozapine Initiation and Titration Chart User Guide 2022](#)
- NSQHS Standards: 1,4,5,6,8

References

1. Australian Medicines Handbook [Internet]. Adelaide (SA): Australian Medicines Handbook Pty Ltd; Clozapine [cited 2023 Aug 15]. Available from: Australian Medicines Handbook.

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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